CariSurg MedTech Pathways Programme - Week 0

Student Information

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Programme: CariSurg MedTech Pathways Cohort 2026
Start Date: May 202
Task: Day 6: Pseudocode for At-Risk Rule (Mercer General ED)

I. INTRODUCTION

The Mercer General Emergency Department (ED) is a regional referral hospital serving approximately 80,000 residents, with additional seasonal population increases due to tourism. The department operates 24/7 under high variability in patient demand, often exceeding 150 patients per day during peak periods.

Due to limited clinical resources and shared treatment spaces, rapid identification of critically unwell patients is essential. This project designs a digital triage system that processes patient vital signs and classifies them into risk categories to support clinical decision-making during triage.

The system is designed as a rule-based decision support tool that assists nurses in prioritising patients based on physiological instability.

II. PATIENT DATA INPUTS

The system uses the following clinical vital signs:

- Pulse Rate (beats per minute)
- Oxygen Saturation (SpO_2 %)
- Temperature (°C)

These are standard emergency triage measurements used to assess cardiovascular, respiratory, and infectious status.

III. DIGITAL TRIAGE PSEUDOCODE

START

Collect patient vital signs

IF $SpO_2 < 90\%$

→ HIGH RISK (Respiratory Compromise)

ELSE IF Pulse > 130 bpm

→ HIGH RISK (Severe Physiological Instability)

ELSE IF Pulse > 100 bpm AND Temperature > 38°C

→ MODERATE RISK (Possible Infectious Process / Early Sepsis)

ELSE IF Pulse < 60 bpm

→ MODERATE RISK (Possible Cardiac Conduction Abnormality)

ELSE IF Temperature > 38°C

→ MODERATE RISK (Possible Fever / Infection)

ELSE

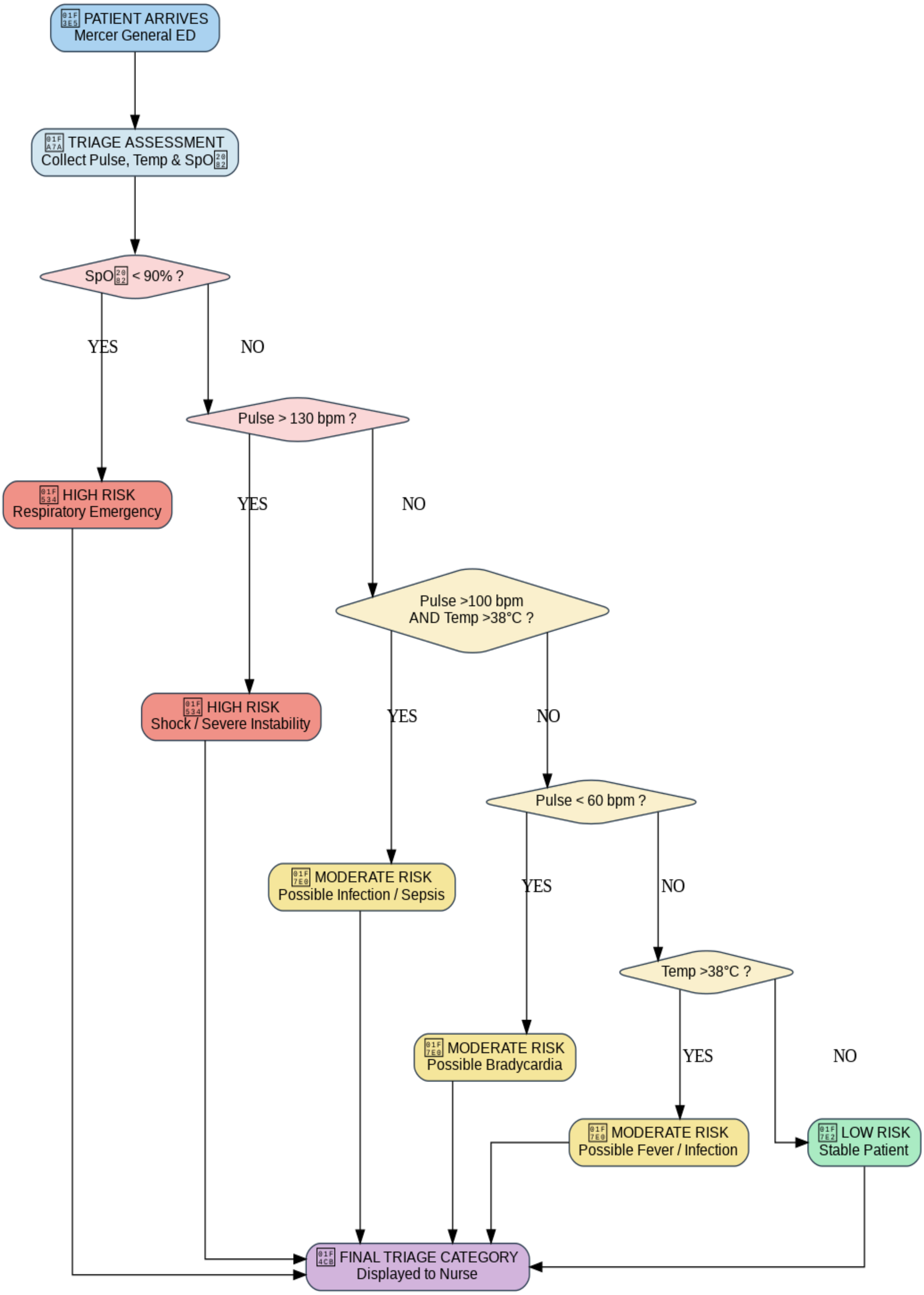
→ LOW RISK (Stable Patient)

END

IV. SYSTEM WORKFLOW

1. Patient arrives at Mercer General ED
2. Triage nurse collects vital signs
3. Digital system receives patient data
4. Data is validated for completeness
5. System evaluates SpO_2 first (critical respiratory marker)
6. Pulse rate is evaluated for cardiac instability
7. Temperature is assessed for infectious processes
8. Combined vital sign patterns are evaluated
9. Patient is assigned a final risk category
10. Result is displayed for clinical prioritisation

Figure 1: Digital Triage Decision Flowchart



VI. CLINICAL LOGIC EXPLANATION

The triage system evaluates multiple physiological parameters in a hierarchical structure. Critical oxygen desaturation is prioritised first due to its direct association with respiratory failure.

Subsequent evaluation of pulse rate identifies cardiovascular instability, while combined pulse and temperature abnormalities assist in early detection of systemic infection or sepsis.

Bradycardia and isolated fever are included as moderate risk indicators to ensure early escalation of patients who may deteriorate.

The system is intentionally rule-based to ensure interpretability, speed, and consistency in emergency environments.

VII. CLINICAL JUSTIFICATION OF THRESHOLDS

SpO_2 < 90% → severe hypoxia requiring immediate intervention

Pulse > 130 bpm → possible shock, sepsis, or cardiac instability

Pulse > 100 bpm + Temp > 38°C → early infectious process or sepsis risk

Pulse < 60 bpm → possible conduction abnormality or medication effect

Temp > 38°C → febrile illness or inflammatory response

VIII. FINAL SUMMARY

This digital triage model demonstrates how emergency department vital signs can be structured into a decision-making algorithm. The system improves consistency in patient prioritisation and supports early detection of clinical deterioration in a high-volume emergency environment.